Enter the Metaverse: Android apps on Meta Horizon OS

START Boot Camp Session 2

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Introduction

This document introduces how to deploy Android applications on the Meta Quest platform, covering the various hardware specifications and user experiences available on the headset. It also introduces the Meta Spatial SDK, which allows developers to adapt existing Android apps or create new ones using their current skills. The guide aims to streamline the development process, ensuring that developers can use Meta Quest's features to improve their apps and user engagement.

Learning Objectives

By reading this guide, you will:

- Understand the key differences between 2D Android development and the Meta Spatial SDK.
- Learn how developing for Meta Quest connects you to a growing XR audience, creating opportunities for user engagement and market growth.
- Build confidence in transforming your mobile app into a mixed-reality experience. Use your existing skills to add spatial features that enhance interactivity and user appeal.

Introduction to Mixed Reality with Meta Quest

Meta Quest was initially developed as a VR headset but has evolved into a platform for mixed reality (MR). This new phase focuses on merging the physical world with the digital realm, starting with familiar surroundings to enhance user comfort and reduce motion sickness.

This allows users to ease into the experience and gradually choose their level of immersion, making mixed reality a great entry point for virtual experiences. Additionally, this approach not only eases the initial experience but also offers a scalable journey into more immersive VR environments.

With this high-level understanding, let's explore the Meta Quest devices and their features.

Meta Quest 3 Devices

The Meta Quest 3 family of devices is designed to integrate seamlessly into mixed reality. Each device operates as a standalone headset with its own operating system and applications, removing the need for external batteries, computers, or cameras.

The headsets feature built-in cameras that track your position in physical space and enable hand tracking for interactive experiences. With no cables required, these devices allow unrestricted movement, enhancing user freedom and immersion

In 2024, Meta introduced the Meta Quest 3 and the Meta Quest 3s. While both devices deliver a consistent experience, Meta Quest 3, the more premium device, has a higher field of view, whereas Meta Quest 3s has approximately 20% fewer pixels per degree. Both Meta Quest 3 devices deliver unique capabilities enhanced by advanced technology for a superior mixed-reality experience:

Meta Quest 3s

Meta Quest 3

- **Processor:** Powered by the Snapdragon XR2 Gen 2 processor, the devices deliver crisp visuals, fast load times, and smooth performance for immersive, high-quality apps.
- **Visuals and Space Awareness:** Full-color, high-resolution passthrough from front-facing cameras blends digital elements with the physical world, enhancing mixed-reality experiences.
- **Controller Precision:** Touch Plus controllers offer precise tracking, ideal for fast-paced applications requiring accurate input.
- **Interaction Flexibility:** Supports hand tracking for intuitive, controller-free interactions, while Bluetooth support allows for specialized use cases.

Headset

Mixed Reality

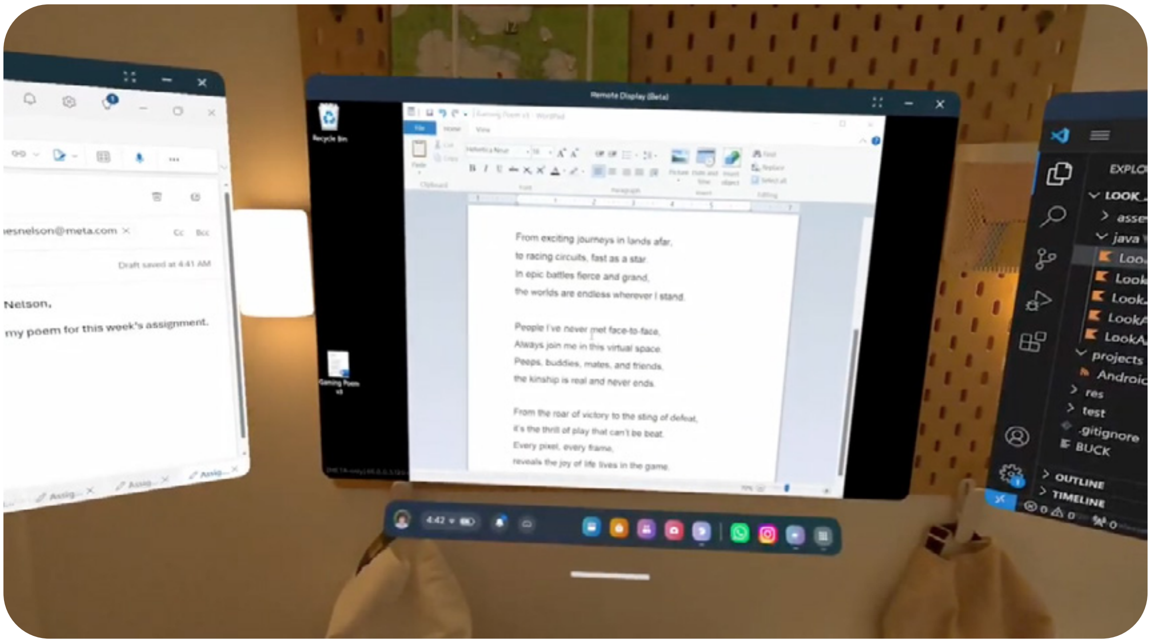
Input

Exploring 2D Apps Inside the Meta Quest Headset

Multi-screen Workstation

Transform your laptop into a powerful desktop environment with the free Meta Quest Remote Desktop feature:

- **Remote Desktop Capabilities:** Connect to a Windows 11 PC with a single click, bringing all your applications into your headset for easy access.
- **Expanded Screen Real Estate:** Access up to three high-resolution virtual displays, creating a multi-screen desktop setup for enhanced productivity.
- **Versatile and Wireless:** Use this feature wirelessly on both Mac and PC, enabling seamless multitasking and detailed work without physical connections.



Lie Back and Relax

Enjoy ultimate comfort with Lie-down mode, designed for seamless device use while reclining:

- **Lie-down Mode:** Continue using your Meta Quest 3 device comfortably while lying down, with automatic re-centering that adjusts your view for optimal interaction.
- **Ideal for Entertainment:** Enjoy enhanced comfort while browsing or watching TV shows and movies.
- **Interactive Control:** Use your hands to control panel placement and playback for a seamless and intuitive experience.

Multi-screen Entertainment

Transform your entertainment experience with multi-screen functionality, offering the power of multiple displays:

- **Versatile Panel Use:** Open up to six panels simultaneously to manage social media, messages, and websites, perfect for multitasking or staying entertained during other activities.
- **Flexible Screen Positioning:** Arrange panels in spatial configurations to suit your preferences, creating a personalized viewing experience.
- **Spatial Audio Features:** Use spatial audio to distinguish between sounds from different applications, improving focus and enhancing multitasking.

Big Screen Video

Theater mode enhances your viewing experience by projecting content onto a large, immersive screen:

- **Immersive Big Screen Display:** Project content onto a large screen for an enhanced, immersive viewing experience. Adjust the screen size to suit your preferences for maximum impact.
- **Customizable Viewing Environment:** Modify ambient lighting to improve visibility and create a more comfortable experience.
- **Enhanced Audio Experience:** Enjoy rich sound quality with Dolby Atmos audio, available for supported apps and content.
- **Available On Any Panel:** Theater mode is accessible from any panel, ready to enhance your multimedia experience.

Xbox Gaming

Unlock a new dimension of gaming with pairing Xbox Cloud gaming with the Meta Quest 3 device's Theater mode:

- **Massive Game Screen:** Experience Xbox Cloud gaming and Theater mode together, simulating a cinematic experience for your favorite games.
- **Simple Connectivity:** Pair a compatible controller via Bluetooth and launch the Xbox Cloud Gaming app to access a wide selection of cloud-based games.
- **Android Compatibility:** Discover how well Android standards integrate with the VR environment, providing a seamless and smooth gaming experience.

Play Your Music

Customize your gaming soundtrack with background media and playback controls:

- **Personalized Soundtracks:** Play your favorite music during gaming to replace or complement the default game audio.
- **Easy Media Control:** Use background media and playback controls to manage your music without disrupting gameplay.
- **Media Session App Compatibility:** Works automatically with any app using a media session, requiring no additional programming.

Spotify (2D App) - Coming soon to Meta Horizon OS

Stay Connected

Discover enhanced connectivity features designed to enrich your experience across various settings and applications:

- **Travel Mode:** Watch downloaded movies or connect to in-flight entertainment systems using your Meta Quest 3. Travel Mode delivers a big-screen experience with added privacy, making it ideal for use on planes or trains.

Travel mode is available on trains and planes.

- **Seamless Multitasking:** Stay engaged with your applications while participating in games or virtual meetings. This feature ensures uninterrupted connectivity within immersive environments.

Seamless multitasking feature is currently in experimental

- **Remote Casting:** Use Android's Media Casting API to share your view with others outside the headset. This feature is perfect for collaborative gaming, remote assistance, or sharing experiences, offering functionality similar to screen-sharing during a video call. It is accessible to any app that integrates the API

Share VR experiences with friends using Android's Media Casting API.

Engagement with browser and 2D apps

Based on Meta internal data, more and more users are engaging with 2D apps on Meta Quest. This data strongly indicates that users want both types of apps, and that there is an opportunity to offer great content for users to discover.

30%
of Meta Quest monthly active users open at least one app on top of their games.[1]

>60%
of Meta Quest monthly active users engage with browser and 2D apps on Meta Quest 3.[2]

Source: 1. Meta internal data, Q3 2024. 2. Meta internal data, 2024.

Bringing Android apps to Meta Horizon OS

This section explores development for Meta Horizon OS, the core operating system powering Meta Quest devices.

Android on Meta Quest

Meta Quest devices run on Meta Horizon OS, a branch of the Android Open Source Project. This means Android apps that run on mobile devices can easily run on Meta Quest.

Sideloading our shell manages how the app operates in mixed reality by:

- **Coordinating Panel Placement:** Ensures panels are positioned appropriately within the virtual environment
- **Projecting Spatial Audio:** Aligns sounds with their correct locations around the user for an immersive experience.
- **Translating Input Events:** Converts user interactions into basic scroll and tap actions.

Developing for Meta Quest is similar to building for any other Android device, allowing you to continue using familiar tools and workflows, such as:

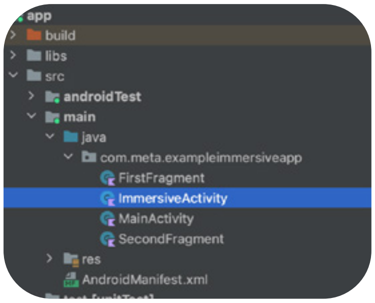
- **Kotlin-based Apps:** Use Jetpack Compose for UI composition.
- **Java-based Apps:** Port legacy applications using Java-based views.
- **Cross-platform Frameworks:** Build for multiple platforms using a framework like React Native.

Our shell simplifies event translation into basic tap and scroll actions, eliminating the need for specialized input processing. Activate development mode via the Meta mobile app and register as a developer on our website to begin sideloading and testing your app immediately.

Building for the future on Meta Horizon OS

Meta Horizon OS streamlines the development process, making it intuitive to create mixed-reality applications:

- **Familiar environment:** Use tools you already know, such as Android Studio or other integrated development environments (IDEs), for seamless integration with Meta Horizon OS.
- **More devices:** Developing for Meta Quest feels familiar, even with its new form factor. Treat it as another device target in your existing development workflows.
- **Increased reach:** Recent updates to the Meta Quest Store provide developers with a fresh platform to promote mobile apps and connect with new audiences seeking unique experiences.

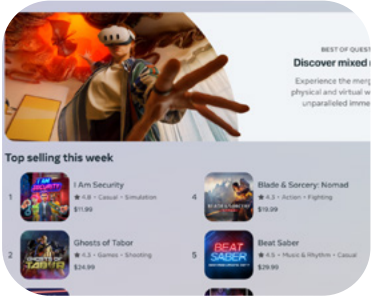


Familiar environment

Create immersive experiences using familiar skills and frameworks.

More devices

Developing for Meta Quest remains familiar while introducing a new device form factor.



Increased reach

Meta Horizon Store provides an exciting platform to showcase your app.

2D apps on Meta Quest

The Meta Quest Store offers a growing variety of 2D apps, leveraging the device's unique capabilities to meet diverse user needs:

- **Diverse App Categories:** Explore a range of 2D apps from partners, focusing on gaming, entertainment, productivity, and social interactions.
- **Innovative Updates:** Recent relaunches of Facebook and Instagram, rebuilt with React Native, optimize performance and usability for larger screens and mixed-reality environments.
- **Evolving Platform:** The Meta Quest Store continues to expand, introducing new apps and possibilities to enhance user experiences and broaden use cases.

The Meta Quest Store unlocks new opportunities, with innovative apps launching regularly.

Continuous Meta Horizon OS software updates

Meta Horizon OS receives frequent updates to enhance user and developer experiences through a dedicated monthly feature release schedule. These updates aim to improve the Android experience on Meta Quest, with each release incorporating feedback and responding to user needs promptly.

Here is an example of a six-month 2024 roadmap:

May v65	June v66	July v67	August v68	September v69
- Spatial videos and panoramas - Travel mode	- Background audio - Media control bar	- Swipe typing - Immersive media continuity - Theater view - Movable panels for multitasking	- Improved performance for 2D apps - Controller pairing in-headset - Audio balancing - Keyboard improvements	- Seamless multitasking - Spatial panel audio - Window layout - Bluetooth quick pairing - Stylus support

Meta Spatial SDK

Traditionally, adding immersive functionality to apps required using a game engine, a process that often posed a steep learning curve for developers unfamiliar with immersive environment design. Moreover, integration with a game engine frequently demanded a complete overhaul of core user flows to conform to its architecture, significantly increasing the complexity and maintenance of codebases.

To address these challenges and make immersive development more accessible to Android developers, Meta introduced the Meta Spatial SDK. This toolkit is designed to simplify the incorporation of immersive elements into apps without requiring expertise in new development environments.

- **Streamlined Development Process:** Meta Spatial SDK enables Android developers to enhance their apps with immersive functionality using a Kotlin-based toolkit. It provides specialized APIs and built-in features for seamless integration with Meta Quest.
- **Ease of Use:** Whether you're upgrading an existing mobile app with spatial functionality or creating entirely new mixed-reality applications, the Meta Spatial SDK is built to simplify the process.
- **Seamless Workflow Integration:** Designed to integrate smoothly into existing development workflows, the SDK reduces the need for extensive codebase modifications, easing the transition into mixed-reality app development.

By leveraging the Meta Spatial SDK, developers can create immersive experiences efficiently, making it an ideal tool for bringing innovative mixed-reality applications to life.

Why Meta Spatial SDK?

The Meta Spatial SDK was designed with Android developers in mind, offering a familiar, efficient, and additive approach to building immersive experiences.

Familiar

Looks and feels very familiar to mobile developers.

Easy

Simplifies Meta Horizon OS development with intuitive APIs.

Fast

Build on top of existing engineering stack and infrastructure.

Additive

Enables developers to leverage the wide ecosystem of libraries and tools.

- **Familiar:** Developing with the Meta Spatial SDK feels like building 2D apps, leveraging the tools and workflows Android developers already know.
- **Easy:** The SDK includes ready-to-use libraries and intuitive APIs, enabling developers to quickly create spatial experiences without steep learning curves.
- **Fast:** Extend your existing tech stack and infrastructure by reusing or adapting 2D user flows for mixed-reality scenes, accelerating development timelines.
- **Additive:** The SDK integrates seamlessly with the Android ecosystem, allowing developers to leverage libraries for UI composition or advanced features like Generative AI. This flexibility enables developers to enhance apps incrementally as needed.

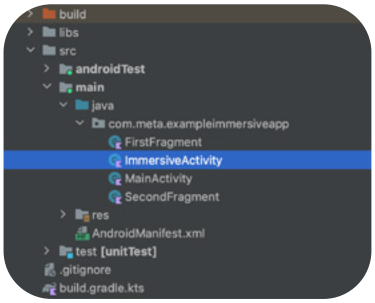
Project Setup

Adding an immersive activity to your project is as straightforward as adding any other activity. By inheriting from a helper activity, developers can efficiently initialize and manage virtual scenes. To enrich your app, simply add objects through code.

For example, to create a simple scene with an interactable red box:

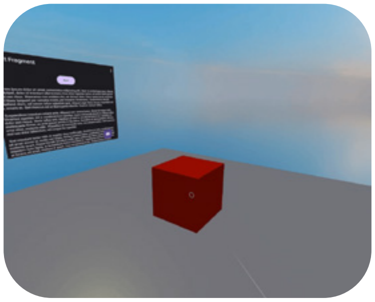
- Set up the object with a 3D model, defining its size, color, and position.
- When the app is launched, users are transported directly into the virtual environment you've created—bypassing traditional panel loading.

This approach immerses users entirely in the experience you've designed, ensuring a seamless and engaging interaction.



Familiar integrated developer environment

Kotlin



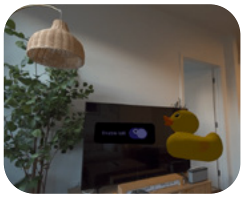
Meta Spatial SDK scene

Meta Spatial SDK capabilities

The Meta Spatial SDK provides a range of built-in capabilities to simplify the creation of immersive mixed-reality experiences:

- Render high-quality 3D models within your scenes to create visually compelling environments.
- Control the user's environment by loading immersive skyboxes or utilizing passthrough cameras to blend physical surroundings with virtual elements.
- Enable users to interact with objects in the scene using inputs from controllers or hand tracking.
- Use the built-in physics engine to enable realistic collisions and interactions between 3D objects.

Modern graphics



Mixed reality

Controllers & hands

Immersive media

Physics

Spatial Panels

Panels serve as powerful tools for presenting information or facilitating user interactions. They are especially effective for delivering text, images, and UI components in an organized, user-friendly way.

Easily create panels using activities and Android UI frameworks. This integration ensures a seamless user experience and allows users to interact effortlessly with your content.

Example: Integrating a video play into an immersive scene

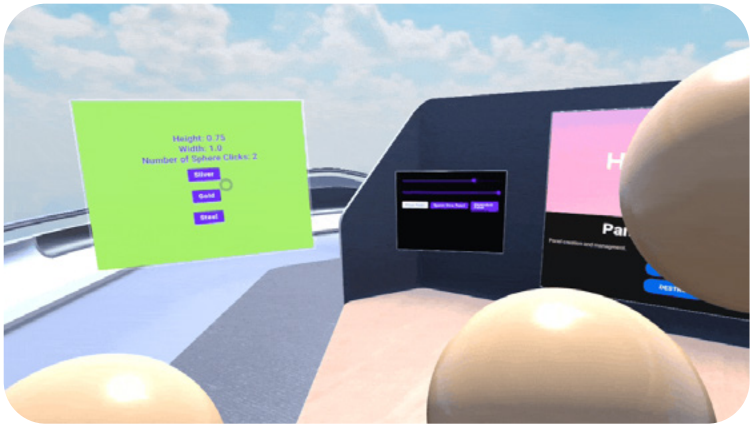
- **Create a Video Player App:** Start by building a basic video player app that functions within an immersive scene.
- **Register the Panel:** Connect your video player to a VideoPlayerActivity by registering the panel and defining specifics such as size, shape, and resolution.
- **Position the Panel:** Spawn the panel at a predetermined position and orientation within the virtual environment.
- **Forward Inputs:** Ensure that user inputs are directed to the underlying activity once the panel is loaded. This setup allows the video player's controls and behaviors to work seamlessly within the immersive environment.

```
override fun registerPanels(): List<PanelRegistration> {
   return listOf(
                 // Register a panel using an Intent
       PanelRegistration(R.id.video_player_panel) {
           panelIntent = Intent().apply {
               setClassName(applicationContext,
VideoPlayerActivity::class.qualifiedName!!)
           }
                         // Configure size and shape
           config {
               height = 2.16f
               width = 3.84f
               layoutHeightInPx = 1080
               layoutWidthInPx = 1920
               layerConfig = QuadLayerConfig()
               panelShader = SceneMaterial.HOLE_PUNCH_SHADER
               alphaMode = AlphaMode.HOLE_PUNCH
           }
       }
   )
}

override fun onSceneReady() {
   super.onSceneReady()
   // Change environment lighting and view position
   scene.setViewOrigin(0.0f, 0.0f, 0.0f)
   scene.setLightingEnvironment(
      ambientColor = Vector3(1.0f, 1.0f, 0.5f),
      sunColor = Vector3(1.0f, 1.0f, 0.5f),
      sunDirection = -Vector3(1.0f, 3.0f, 2.0f),
      environmentIntensity = 0.5f
   )

   // Create Panel at a set position
   Entity.createPanelEntity(
     R.id.video_player_panel,
     Transform(Pose(Vector3(0f, 1.3f, 3.5f), Quaternion(0f, 0f
```

2D to 3D communication



Seamlessly integrate interactions between 2D panels and 3D objects in your scene by following these steps:

- **Utilize Android ViewModels:** Manage and update data between 2D and 3D objects using Android ViewModels. This approach ensures efficient handling of data changes within your scene.
- **Track Interactions:** For example, to monitor interactions with a 3D sphere, update the clickCount in the ViewModel based on trigger events from the controller. This enables real-time updates between 2D and 3D elements.
- **Enable Event-Based Communications:** Implement event-based communication to allow immersive activity functions to be called directly from your 2D code. This supports dynamic interactions within the virtual environment.
- **React to 2D Inputs in 3D Contexts:** Configure your immersive activity to respond to 2D panel inputs. For instance, when a user presses a button on a 2D panel, you can alter properties of 3D objects, such as changing the material of a sphere in the scene.

These steps ensure a dynamic and interactive environment that bridges the gap between 2D UI elements and 3D immersive experiences.

```
Entity.fromSphere(
              .3f,
              Transform(Pose(Vector3(x = x - 1.5f, y = 1f, z = 1f))),
              FlatColorMaterials.TAN,
              Grabbable(),
              FloatingDependentOrb(),
          )
.registerEventListener<ButtonReleaseEventArgs>(ButtonReleaseEventArgs.EVENT_NAME) {
              _,
              eventArgs ->
            if (eventArgs.button == ControllerButton.RightTrigger) {
              PanelViewModel.clickCount.value++
            }
          }
```

```
onClick = {
           SpatialActivityManager.executeOnVrActivity<PanelManagementActivity> {
activity ->
             val q = Query.where { has(FloatingDependentOrb.id) }
             for (entity in q.eval()) {
               entity.setComponent(activity.goldMaterial)
             }
           }
         },
```

Hybrid apps

Hybrid apps combine the accessibility of 2D panels with the immersive depth of 3D applications, offering a versatile framework for user interaction:

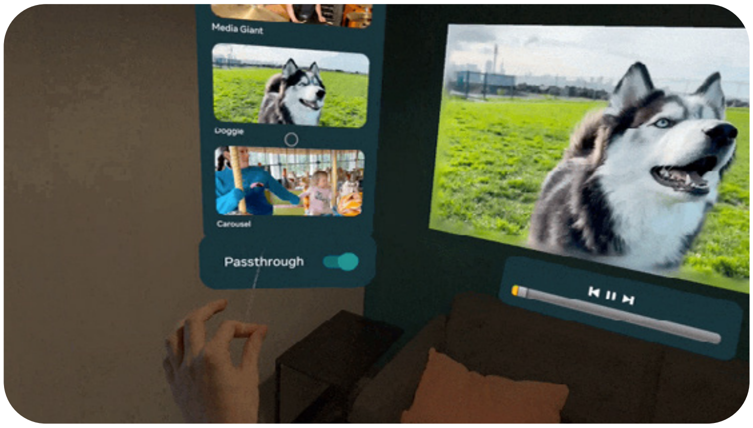
- **Ease of Transition:** Hybrid apps enable seamless transitions between 2D panels, such as a standard browser, and fully immersive 3D apps when the user experience requires greater depth.
- **Versatile Application Use:** For example, a video player can start as a simple panel in the Home environment and expand into a fully immersive 3D scene, adapting to user needs.
- **Flexible User Interaction:** Hybrid apps support both multitasking-friendly 2D views and fully immersive 3D environments within the same application, catering to diverse user preferences and interaction styles.

With a single button press, users can shift from a 2D panel into an immersive scene or return back to the 2D view. This dual approach ensures flexibility and adaptability to a wide range of use cases and preferences.

Immersive and passthrough

Switching between immersive environments and real-world surroundings enriches user interactions while offering versatile options for engagement.

- **Immersive Backgrounds:** Use developer-created immersive skyboxes to craft compelling backgrounds that enhance the depth and engagement of virtual scenes.
- **Real-World Integration:** Leverage passthrough cameras to overlay apps onto the user's physical surroundings, seamlessly merging digital and real environments for augmented reality experiences.
- **Versatile Showcase Options:** The Meta Spatial SDK supports a variety of contexts, from fully virtual scenes to mixed-reality applications, catering to diverse user preferences and use cases.



Modern 3D graphics

Leverage the full spectrum of modern 3D graphics technologies to create visually stunning and realistic experiences.

- **Supports GLTF/GLB models:** Utilizes the GLTF / GLB file formats, which are optimized for efficient, high-quality 3D rendering.
- **Advanced Rendering Techniques:** Features Physically-Based Rendering (PBR) and Image-Based Lighting for lifelike lighting effects. These techniques ensure realistic interactions between light and objects in your scenes, significantly enhancing visual quality.
- **Customization Options:** While offering robust support for standard shaders for realistic effects, the Meta Spatial SDK also provides experimental support for custom shaders. This enables developers to tailor the appearance of their scenes to meet specific artistic visions.

Meta Spatial Editor

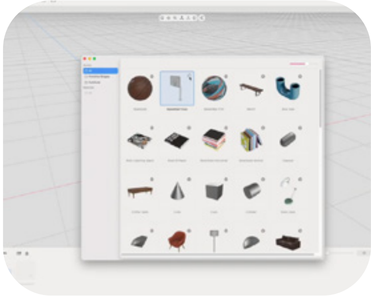
Transform your asset management and scene creation process with the Meta Spatial Editor, a powerful desktop editor tool designed to simplify the development of immersive experiences for Meta Horizon OS. Here's how it enhances your workflow:

- **Efficient Asset Management:** Import, organize, and transform your assets into visual compositions. Export them directly into the Meta Spatial SDK to bring your immersive scenes to life.
- **Intuitive Scene Composition:** Easily compose scenes using models, panels, and skyboxes in a user-friendly environment.
- **Seamless Export to SDK:** Export your composed scene for use within an app developed with the Meta Spatial SDK.
- **Development Efficiency:** Save time and fine-tune scenes during the development phase, ensuring the final product is visually and functionally aligned with project goals.

This tool is essential for developers looking to efficiently create and iterate on immersive environments without the tedium of coding each element by hand.

In the example below, a basketball court is set up using the following steps:

- **Import Assets:** Import the required assets, such as the hoop, basketball, scoreboards, and other elements.
- **Compose the Scene:** Set up the basketball court, designating the basketball as grabbable.
- **Export Composition:** Export the completed scene composition for use in your app.
- **Specify Scene in App:** In the application, specify the exported composition to load and run within the app.
- **Separate Logic from Scene Construction:** Keep scene construction separate from application logic to ensure modular and maintainable development.



Meta Quest TV is powered by Spatial SDK

The Meta Spatial SDK supports popular immersive media formats, such as 180° and 360° photos and videos, enabling developers to create rich, panoramic, and stereoscopic experiences. Here's how you can leverage this capability:

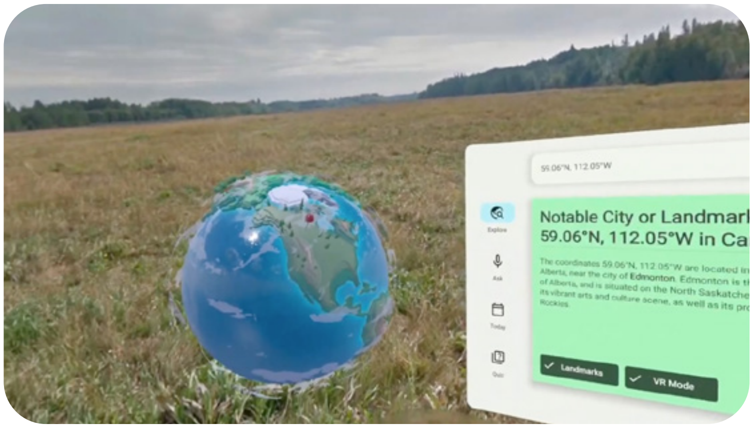
- **Diverse Media Formats:** Render high-quality videos in formats such as 180° and 360° panoramic videos with stereoscopic 3D. The Meta Quest TV app serves as an example of how to utilize the SDK for these capabilities, demonstrating its potential in immersive media applications.
- **Accessible Developer Resources:** Explore the media viewer showcase on GitHub, which provides practical insights, source code, and a valuable starting point for building your own immersive media applications.

Open Source Developer Showcase Apps

We developed showcase apps to demonstrate how the Meta Spatial SDK can be used to build Productivity and Educational apps:

- **Focus:** A productivity app allowing users to set up projects, track tasks, create notes pinned within the environment, and use timers to manage work.

- **Geo Voyage:** An educational app that uses Generative AI to provide trivia about locations worldwide, paired with 360° views of these locations for an immersive learning experience.



These showcase apps are open source and available on GitHub. They serve as excellent references for developers looking to build their own spatial apps.

Key Takeaways

Start with the app and skills you have

Begin with your existing app and toolchain. Prototype in 2D if you're not ready to spatialize yet or want to test concepts before committing.

Try a few new things

Experiment with features like theater mode or spatial layouts with panels side-by-side. Test how things feel in the headset before finalizing decisions.

Go play!

There are no one-way technical doors with Spatial SDK. Its adherence to Android standards ensures your existing app works as-is, allowing you to switch between 2D and spatial features seamlessly. This flexibility enables experimentation while keeping the focus on improving user experiences.